Sl.No.	Lessons/Topics to be covered	Book Reference (sections)
1	Introduction to the course and Discussion on Course Outcomes (COs), Program Outcomes (POs) and their mapping, Lesson Plan, Grading and Evaluation Pattern.	Course Handbook
2	Introduction to Computers and Python: Introduction, Hardware and Software, Data Hierarchy, Machine Languages, Assembly Languages and High-Level Languages, Introduction to Object Technology, Operating Systems, Python	PH_1.1-1.7 (pg.1-17)
3	It's the Libraries!, Other Popular Programming Languages, Test-Drive: Using IPython and Jupyter Notebooks	PH_1.8-1.10 (pg.18-29)
4	Internet and World Wide Web, Software Technologies, How Big is Big Data?, Intro to Data Science: Case Study– A Big-Data Mobile Application	PH_1.11-1.14 (pg.29-48)
5	Introduction to Python Programming: Introduction, Variables and Assignment Statements, Arithmetic, Function Print and an Intro to Single- and Double-Quoted Strings, Triple Quoted Strings	PH_2.1-2.5 (pg.49-59)
6	Getting Input from the User, Decision Making: The if Statement and Comparison Operators	PH_2.6-2.7 (pg.59-66)
7	Objects and Dynamic Typing, Intro to Data Science: Basic Descriptive Statistics	PH_2.8-2.9 (pg.66-72)
8	Control Statements and Program Development: Introduction, Algorithms, Pseudocode, Control Statements, if Statement, if...else and if...elif...else Statements	PH_3.1-3.6 (pg.74-85)
9	Iteration (for and while Statements) with examples	PH_3.7-3.8 (pg.85-89)
10	Augmented Assignments, Program Development: Sequence-Controlled Repetition, Program Development: Sentinel-Controlled Repetition, Program Development: Nested Control Statements	PH_3.9-3.12 (pg.89-100)

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)
11	Built-In Function range: A Deeper Look, Using Type Decimal for Monetary Amounts, break and continue Statements	PH_3.13-3.15 (pg.101-106)
12	Boolean Operators and, or and not, Intro to Data Science: Measures of Central Tendency–Mean, Median and Mode	PH_3.16-3.17 (pg.106-117)
13	Functions: Introduction, Defining Functions, Functions with Multiple Parameters, Random-Number Generation	PH_4.1-4.4 (pg.120-128)
14	Case Study: A Game of Chance, Python Standard Library, math Module Functions	PH_4.5-4.7 (pg.128-133)
15	Using IPython Tab Completion for Discovery, Default Parameter Values, Keyword Arguments, Methods: Functions That Belong to Objects, Scope Rule	PH_4.8-4.13 (pg.133-140)
16	import: A Deeper Look, Passing Arguments to Functions: A Deeper Look, Passing Arguments to Functions: A Deeper Look, Functional-Style Programming, Intro to Data Science: Measures of Dispersion	PH_4.14-4.18 (pg.140-154)
17	Sequences: Lists and Tuples: Introduction, Lists, Tuples, Unpacking Sequences	PH_5.1-5.4 (pg.156-166)
18	Sequence Slicing, del Statement, Passing Lists to Functions, Sorting Lists	PH_5.5-5.8 (pg.166-173)
19	Searching Sequences, Other List Methods, Simulating Stacks with Lists, List Comprehensions, Generator Expressions, Filter, Map and Reduce	PH_5.9-5.12 (pg.173-181)
20	Generator Expressions, Filter, Map and Reduce, Other Sequence Processing Functions	PH_5.13-5.15 (pg.181-187)

Sl.No.	Lessons/Topics to be covered	Book Reference (sections)
21	Two-Dimensional Lists, Intro to Data Science: Simulation and Static Visualizations	PH_5.16-5.17 (pg.187-208)
22	Dictionaries and Sets: Introduction, Dictionaries, Example: Word Counts	PH_6.1-6.2.7 (pg.210-220)
23	Dictionary Method update, Sets	PH_6.2.8-6.3 (pg.220-228)
24	Intro to Data Science: Dynamic Visualizations	PH_6.4 (pg.228-238)
25	Array Oriented Programming with NumPy: Introduction, Creating arrays from Existing Data, array Attributes, Filling arrays with Specific Values, Creating arrays from Ranges	PH_7.1-7.5 (pg.240-246)
26	List vs. array Performance: Introducing %timeit, array Operators, NumPy Calculation Methods	PH_7.6-7.8 (pg.246-252)
27	Universal Functions, Indexing and Slicing, Views: Shallow Copies, Deep Copies	PH_7.9-7.12 (pg.252-258)
28	Reshaping and Transposing, Intro to Data Science: pandas Series and DataFrames	PH_7.13-7.14 (pg.259-282)
29	Introduction, Formatting Strings, Concatenating and Repeating Strings	PH_8.1-8.3 (pg.284-290)

Sl.No.	Lessons/Topics to be covered
30	Stripping Whitespace from Strings, Changing Character Case, Comparison Operators for Strings, Searching for Substrings, Replacing Substrings, Splitting and Joining Strings
31	Characters and Character-Testing Methods, Raw Strings, Introduction to Regular Expressions
32	Intro to Data Science: Pandas, Regular Expressions and Data Munging
33	Files and Exception: Introduction, Files, Text-File Processing, Updating Text Files
34	Serialization with JSON, Focus on Security: pickle Serialization and Deserialization, Additional Notes Regarding Files
35	Handling Exceptions, finally Clause, Explicitly Raising an Exception, (Optional) Stack Unwinding and Tracebacks
36	Intro to Data Science: Working with CSV Files